

Assignment 3

Frontend implementation

Overview

In this assignment, you will begin implementing the frontend of your app based on the design you created in Figma. Your implementation should use mock data stored in memory and focus on the basic functionalities of adding a new entity and displaying all entities. Make sure to include data validation to ensure the integrity of the inputs. Optionally, you can extend your implementation to support the full range of CRUD operations (edit, delete) as well as filter and sort capabilities. Incorporating unit tests or following a test-driven development approach is a nice-to-have for extra credit.

Objectives

- **Frontend implementation:**
Translate your Figma design into a functional frontend interface.
- **Mock data handling:**
Use in-memory mock data to simulate backend responses.
- **Core functionalities:**
Implement the "add" and "show all" features for your app's entities.
- **Data validation:**
Ensure that user inputs are validated properly to maintain data quality.
- **Optional enhancements:**
Expand your application with the remaining CRUD operations (update and delete) and add filter/sort functionality.
- **Unit tests/TDD:**
If possible, integrate unit tests or adopt a test-driven development process to ensure code quality.

Assignment tasks

1. Set up your frontend environment:

- Configure your project using your chosen framework.
- Ensure that your development environment supports live reloading for a smooth development experience.

2. Implement the UI based on your Figma design:

- Recreate the frontend layout and components as designed in Figma. Ideally you can export the design with Figma plugins directly into your frontend code.
- Use mock data stored in memory to simulate the backend.

3. Implement core functionalities:

- **Add entity:** Create a form that allows users to add a new entity.
 - Include necessary fields as per your design.
 - Implement data validation to check for errors (e.g., required fields, proper data types).
- **Show all entities:** Develop a view that lists all the entities added.
 - Display the list dynamically using the in-memory data.

4. Optional enhancements:

- **Full CRUD operations:** Extend your application to allow updating and deleting entities.
- **Filter and sort:** Implement functionality to filter and sort the list of entities based on various criteria.
- **Unit tests/TDD:** Write unit tests or follow a test-driven development approach to validate your core functionalities.

5. Documentation:

- Provide a README file with clear instructions on how to run and test your application.
- Document any additional features or enhancements you have implemented.

Submission

● **Deadline:**

Upload your project as a zip file (or share a link to your repository) on the Teams assignment by the end of the lab session.

● **Requirements:**

Ensure that your submission includes:

- The source code of your frontend implementation.
- The README file with instructions.

- A brief summary (optional) of any extra functionalities you implemented (e.g., full CRUD, filter/sort, unit tests).

PLEASE DO NOT UPLOAD anything more than the source code and the README.md. If you don't know what files are needed, create a git repo and add the link of the repo to the assignment, with [gitignore](#)